

Profile Summary

Innovative and detail-oriented ECE graduate with strong hands-on experience in Embedded Systems development using C/C++, Embedded C, and Python. Experienced in designing and debugging real-time embedded applications on ESP32, ESP8266, and Raspberry Pi, involving sensor interfacing, communication protocols, and multitasking concepts similar to RTOS. Winner of Smart India Hackathon with proven ability in hardware-software integration, system optimization, and post-deployment support.

Technical Skills

- **Languages & Core:**

C, C++, Python, Embedded C, Arduino IDE

- **Hardware Platforms:**

Raspberry Pi, ESP32, Arduino, Sensors & Actuators

Languages Known

- English
- Tamil

Education

B.E. in Electronics and Communication Engineering (ECE)

November 22 - May 26

Velalar College Of Engineering And Technology

CGPA: 7.72

Experience & Internship

Freelance Embedded & IoT Developer

January 24 - Present

Self-Employed

- Designed, coded, tested, and debugged embedded system software for ESP32, ESP8266, and Raspberry Pi platforms.
- Implemented multitasking logic for sensor acquisition, communication, and alert handling.
- Integrated hardware peripherals using UART, I2C, and SPI protocols.
- Analyzed system performance and optimized power and communication efficiency.
- Provided post-deployment support, documentation, and debugging assistance.

Projects

Sight Aid - AI-Powered Assistive System

December 24 - April 25

Raspberry Pi | YOLOv5 | LIDAR | TTS | Python

- Developed a smart helmet for visually impaired individuals integrating camera, LIDAR, and ultrasonic sensors.
- Implemented real-time object detection, obstacle alerting, and navigation via voice feedback in Tamil and English.
- Selected in IIT PALS Innowah Hackathon and under product development phase.

Early Decomposition Of Fecal Matter

December 23

Embedded Systems | IoT Monitoring | Sensor Automation

- Designed a rapid composting system using controlled biochemical decomposition.
- Integrated sensor-based feedback and IoT monitoring for process optimization.
- Presented to government bodies and received recognition as SIH1425 winners.

Smart Bus Tracker – Real-Time Location and Seat Monitoring

May 25

Arduino | ESP8266 | GPS (NEO-6M) | FSR402 | Flask | HTML/CSS/JS

- Built an IoT-based system for real-time bus tracking and seat monitoring via a web dashboard.
- Integrated GPS and FSR sensors with Arduino-ESP8266 to send live data to a Flask server.
- Created an interactive website showing bus routes, live map updates, and seat status.

Certifications

Embedded System

December 24

NextGen

Core Java

October 25

Udemy

Full Stack Development (workshop)

January 25

Finest Coder

Web Development

September 25

Rinex

Achievements

Smart India Hackathon 2023

December 25

Winner

YUKTI Innovation Challenge

March 25

Finalist

AICTE & MIC Innovation Cell

September 25

Alumni Member

PALS ACDC 2024-25

April 25

Winners